import beyond the timeline indicated against them. The government is encouraging foreign strategic partnerships and is bringing about de-licensing, de-regulation, export promotion, and foreign investment liberalisation to ensure ease of doing business.

Ministry of Defence has formulated a draft Defence Production and Export Promotion Policy 2020 (DPEPP) to position India amongst the leading countries of the world in defence and aerospace sectors. Two defence industrial corridors are being set up in Uttar Pradesh and Tamil Nadu with a target investment of $2.85 billion by 2024. The space domain, which was dominated previously by the state-run space organisations, is now co-inhabited with strong collaborative linkages between the private and public sector.

ISRO and DRDO are continuously encouraging private sector to participate in their programmes. For example, launch vehicles are being developed with public-private partnerships; GSLV Mk-III has several components manufactured by private entities. ISRO is implementing a five-step programme in partnership with the private sector for making available the newly developed technologies for commercial use.

Draft Space Activities Bill stipulates a major role for the non-governmental and private sector in space domain. To foster innovation and technology development, MSMEs, start-ups, individual innovators, R&D institutes, and academia are being provided resources, including financial support. A technology development fund has been created under DRDO to promote self-reliance in defence technology.

There is emphasis that the effort should be indigenous and all vertical integration takes place within India with minimal dependence on foreign capabilities throughout the supply chain. To reduce dependency on external sources several initiatives have been implemented by the government. As per the government, public-private partnership (PPP) can spark a revolution in defence production, just like the Green Revolution and White Revolution.

Fig. 9: Hundreds of millions of pieces of space junk orbit the Earth daily (Source: https://news.mit.edu)

The government has taken many steps which not only fulfil the requirements of the private industry but also create sustainable and long-term linkages with foreign original equipment manufacturers to meet global demands. The private sector needs to leverage opportunities and focus on areas like space, cyberspace, and artificial intelligence to build weapon systems.

The road ahead

Today, militaries all over the world rely on satellites for command and control, communication, navigation, monitoring early warning and operating space based missile defence systems. Superpower politics and the rising-power dynamics of the 21st century have generated an arms race in space, resulting in a shift from the mere militarisation of space to a trend towards actual weaponisation of space. Countries like the USA, Russia, China, and North Korea have demonstrated nuclear and conventional capability of destroying the adversary's assets in space.

Even as India continues with a policy of non-weaponisation and peaceful uses of outer space, the growing trend towards weaponisation in neighbourhood and the larger global context are influencing India's orientation and thinking. It has become necessary to delineate space programme into civilian and military components with clear-cut institutional architecture and better financial allocation.

From its humble origin in the 1960s, India's space programme has come a long way. Today, the country stands among major world players in satellite launching technology. The private sector has been contributing at a steady pace to meet the increasing demands. Several companies, such as Exseed Space, Pixxel, and Azista, have been providing space assets as well as the ground infrastructure for satellites.

The 2017 Draft Space Activities Bill provides a pathway to cutting out a lot of the bureaucratic red tape. The private sector participation has the potential to boost research and development and innovation while reducing the burden on public spending. This will be particularly significant for an invigorated military space programme to accelerate technology and capability development.

ASAT capabilities demand several technologies related to space based sensors, synthetic aperture radars, electronics, a reliable navigation system, guidance and control, and global positioning systems. Several diverse kinds of sensors—such as infrared sensors, optical sensors, electronic-optical sensors, and magnetic sensors—are important to monitor, detect, and help in sensing the events. If India is able to build this electronics expertise, its global position in terms of anti-missile programme will move up dramatically. EFY